

Load Interconnection System Impact Study Report

For Queue Project Q16-0630

Speed 138 kV

July 04, 2017

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0630. The Interconnection Customer proposes a 211.5 MW large load (data center) facility interconnecting at the Speed 138 kV substation in the Kentucky area. The request was received on June 16, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 307 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 11 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$6,582,000.

3. Network Impacts

3.1 Load Deliverability Analysis

1. Facility: RAPIDS ENERGY CTR / CASINO 138kV
Contingency: SKEENA / PIN HOOK 138kV
Loading: 162.0% (Rating: 587 MVA)
DFAX: 6.2%
Pre-project loading: 159.8%
2. Facility: BARNABIE CABLE STA / CASINO 138kV
Contingency: SKEENA / PIN HOOK 138kV Line
Loading: 160.5% (Rating: 599 MVA)
DFAX: 6.2%
Pre-project loading: 158.3%
3. Facility: ANSON NORTH / PIN HOOK 138kV L
Contingency: BARNABIE CABLE STA / CASINO 138kV
Loading: 152.1% (Rating: 471 MVA)
DFAX: 7.0%
Pre-project loading: 149.0%
4. Facility: SKEENA / PIN HOOK 138kV
Contingency: BARNABIE CABLE STA / CASINO 138kV
Loading: 150.5% (Rating: 483 MVA)
DFAX: 7.0%
Pre-project loading: 147.4%
5. Facility: CAMPBELL JCT / RAPID ENERGY CTR 138kV Line
Contingency: SKEENA / PIN HOOK 138kV
Loading: 141.0% (Rating: 659 MVA)
DFAX: 6.2%
Pre-project loading: 139.0%
6. Facility: THONTOWN / DEWITT 138kV
Contingency: LIVERMORE SUBSTATION / LEE COUNTY SOLID WASTE ENERGY 345kV
Loading: 134.5% (Rating: 259 MVA)

DFAX: 7.8%

Pre-project loading: 128.1%

7. Facility: PRAIRIE HORIZON AGRI ENERGY SUBSTATION / THORNTOWN SUBSTATION 138kV

Contingency: LEE COUNTY SOLID WASTE ENERGY / LIERMORE 345kV

Loading: 131.5% (Rating: 256 MVA)

DFAX: 7.8%

Pre-project loading: 125.0%

8. Facility: CULVER / 51ST ST 138kV line

Contingency: BOZEMAN S / ANSON NORTH 138kV

Loading: 118.0% (Rating: 310 MVA)

DFAX: 5.9%

Pre-project loading: 114.0%

9. Facility: ANSON N / 51ST ST 138kV

Contingency: BOZEMAN SOUTH / ANSON NORTH 138kV

Loading: 115.8% (Rating: 333 MVA)

DFAX: 5.9%

Pre-project loading: 112.1%

10. Facility: SIMPSON TACOMA KRAFT BIOMASS / ZELLWOOD 138kV

Contingency: THORNTOWN / DEWITT 138kV line

Loading: 104.3% (Rating: 257 MVA)

DFAX: 9.4%

Pre-project loading: 96.6%

11. Facility: ZELLWOOD / GREENVILLE AVE 138kV

Contingency: DEWITT / THORNTOWN 138kV Line

Loading: 102.1% (Rating: 256 MVA)

DFAX: 9.4%

Pre-project loading: 94.3%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
CASINO / RAPIDS ENERGY CENTER 138kV	Rebuild line with higher-capacity conductor	\$3,801,000	4% (\$154,000)
BARNABIE CABLE STA / CASINO 138kV	Rebuild line with higher-capacity conductor	\$10,120,000	4% (\$411,000)
ANSON NORTH / PIN HOOK 138kV	Rebuild line with higher-capacity conductor	\$16,501,000	5% (\$790,000)
SKEENA / PIN HOOK 138kV	Rebuild line with higher-capacity conductor	\$7,375,000	5% (\$353,000)
CAMPBELL JCT / RAPID ENERGY CTR 138kV Line	Rebuild line with higher-capacity conductor	\$3,755,000	4% (\$152,000)
THORNTOWN / DEWITT 138kV	Rebuild line with higher-capacity conductor	\$5,360,000	11% (\$603,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
PRAIRIE HORIZON AGRI NERGY / THORNTOWN SUBSTATION 138kV L	Rebuild line with higher-capacity conductor	\$7,241,000	11% (\$814,000)
CULVER / 51ST STREET 138kV	Rebuild line with higher-capacity conductor	\$5,979,000	8% (\$468,000)
ANSON N / 51ST ST 138kV	Rebuild line with higher-capacity conductor	\$19,513,000	8% (\$1,527,000)
SIMPSON TACOMA KRAFT BIOMASS / ZELLWOOD 138kV line	Reconductor line	\$4,052,000	15% (\$615,000)
GREENVILLE AVE / ZELLWOOD 138kV	Reconductor line	\$4,579,000	15% (\$695,000)
	Total allocated to project		\$6,582,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.